Introduction to Embodied AI

Lecture 3 - Robotics II

Prof. He Wang

Logistics

- Assignment 1: to be released this Wednesday night, due 4/4 11:59PM.

论坛	描述	帖子总数	未读帖子	未读对我的回复	参与者总数
Assignment 1 discussion		0	0	0	0

1. **Rotation functions**: You will create a simple toolbox for handling rotations.
2. **Galbot forward kinematics**: Given a galbot 7-dof arm, you need to compute the position and orientation of its end effector.

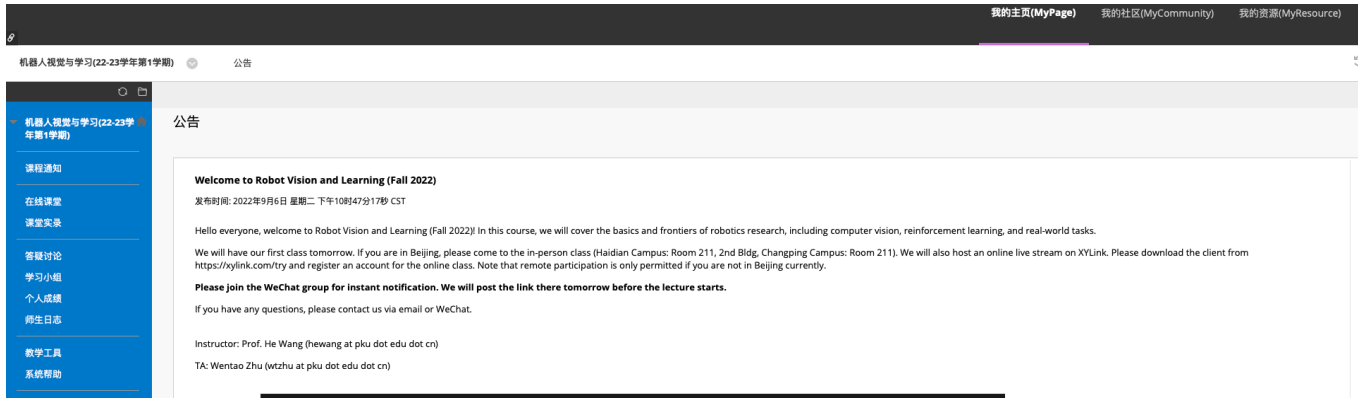
Assignments

- Each assignment will give you more than two weeks to finish.

- Late policy for assignments
  - If 1 day (0 - 24 hours) past the deadline, 10% off
  - If 2 day (24 - 48 hours) past the deadline, 20% off
  - Zero credit if more than 2 days.

Discussion Board

- Internal course website in https://course.pku.edu.cn/
  - Discussion board
    - Assignment 1
    - Assignment 2
    - …



$$\mathbb{SO}(3)$$

Rotations

Rotations are rigid body displacements, *i.e.*, must satisfy two important properties

1. A rotation preserves lengths

$$\|\mathrm{Rp}\| = \|\mathrm{p}\|$$

2. Cross products are preserved by a rotation

$$\mathrm{Rp} \times \mathrm{Rq} = \mathrm{R}(\mathrm{p} \times \mathrm{q})$$

Which implies

$$RR^T = R^T R = I, det(R) = 1$$

For all rotation matrices

Euler Angle is Very Intuitive

$\mathbb{SO}(3)$: The Space of Rotations

- $\mathbb{SO}(n) = \{R \in \mathbb{R}^{n\times n}: det(R) = 1, RR^T = I\}$
- $\mathbb{SO}(n)$: "Special Orthogonal Group"

- "Group": roughly, closed under matrix multiplication
- "Orthogonal": $RR^T = I$
- "Special": $det(R) = 1$

- $\mathbb{SO}(2)$: 2D rotations, 1 DoF
- $\mathbb{SO}(3)$: 3D rotations, 3 DoF

The Issues of Rotation Matrix as Rotation Representation

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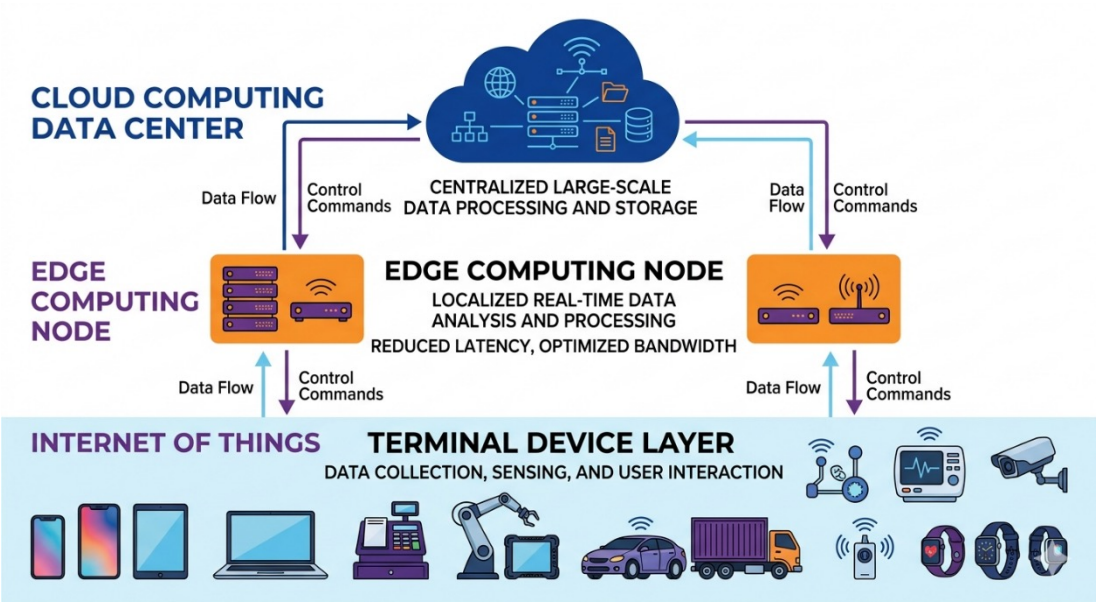
1. Efficiency

Rotation has 3 DoF, however a rotation matrix needs 9 values



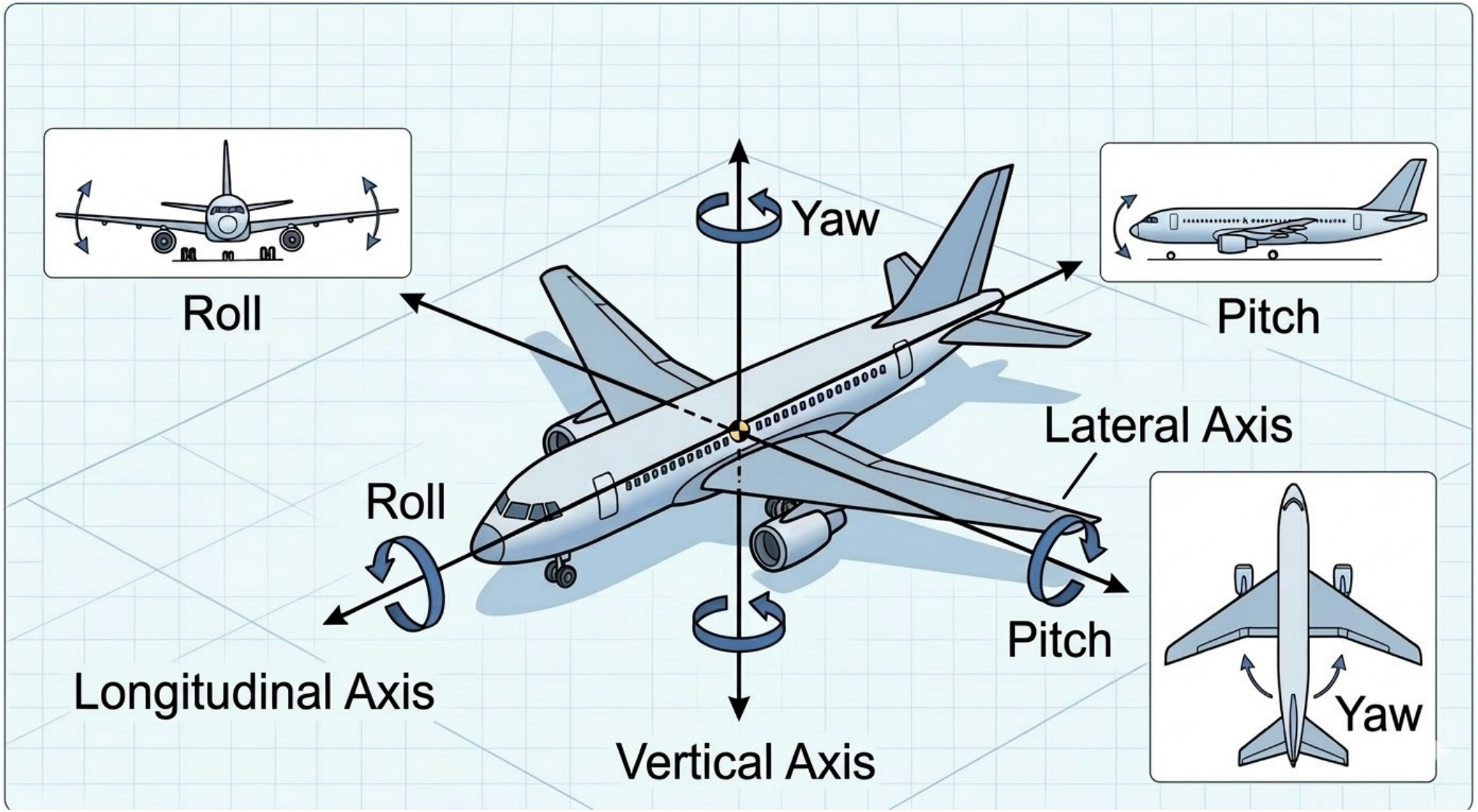
Cloud – Edge – Terminal

NVIDIA Jetson Thor

2. Numerical stability

Maintain orthogonality

Euler Angle


Roll
Yaw
Pitch
Lateral Axis
Roll
Pitch
Longitudinal Axis
Vertical Axis
Yaw

Euler Angle

- The Euler angles are three angles introduced by Leonhard Euler to describe the orientation of a rigid body with respect to a fixed coordinate system.

- Rotation about principal axis is represented as:

$$R_x(\alpha) := \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\alpha & -\sin\alpha \\ 0 & \sin\alpha & \cos\alpha \end{bmatrix}$$

$$R_y(\beta) := \begin{bmatrix} \cos\beta & 0 & \sin\beta \\ 0 & 1 & 0 \\ -\sin\beta & 0 & \cos\beta \end{bmatrix}$$

$$R_z(\gamma) := \begin{bmatrix} \cos\gamma & -\sin\gamma & 0 \\ \sin\gamma & \cos\gamma & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- $R = R_z(\gamma)R_y(\beta)R_x(\alpha)$ for arbitrary rotation (yaw-pitch-row)

Gimbal Lock

- For example: When $\beta = \pi/2$,

$$R = R_z(\gamma)R_y(\pi/2)R_x(\alpha)$$
$$= \begin{bmatrix} 0 & 0 & 1 \\ \sin(\alpha+\gamma) & \cos(\alpha+\gamma) & 0 \\ -\cos(\alpha+\gamma) & \sin(\alpha+\gamma) & 0 \end{bmatrix}$$

since changing α and γ has the same effects, a degree of freedom disappears!

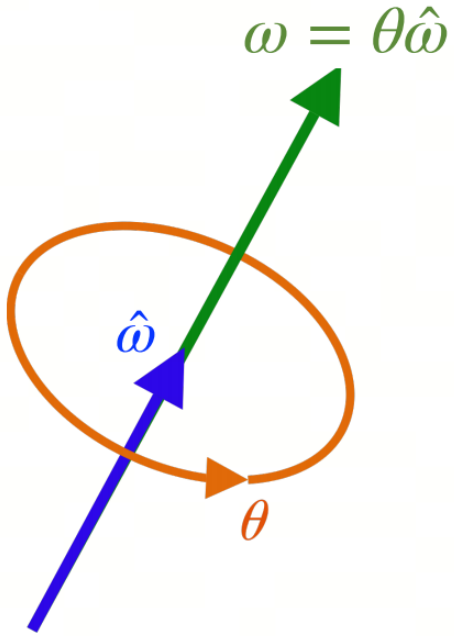
- Not unique!

Summary

- Euler angle can parameterize every rotation and has good interpretability
- It is not a unique representation at some points
- There are some points where not every change in the target space (rotations) can be realized by a change in the source space (Euler angles)

Angle-Axis Parameterization of Rotations

Euler's Theorem



- **Any rotation is equivalent to a rotation about a fixed axis $\hat{\omega} \in \mathbb{R}^3$ ($\|\hat{\omega}\| = 1$) through a positive angle θ**
- $\hat{\omega}$: unit vector of rotation axis
- θ: angle of rotation
- $R \in \mathbb{SO}(3) := \mathrm{Rot}(\hat{\omega}, \theta)$

Given $\hat{\omega}$ and θ, what is $R \in \mathbb{SO}(3)$?

Rodrigues formula

$$\begin{aligned}\mathrm{Rot}(\hat{\omega}, \theta)x &= x + (\sin\theta)\hat{\omega}\times x + (1-\cos\theta)\hat{\omega}\times(\hat{\omega}\times x) \\ &= \{I + [\hat{\omega}]\sin\theta + [\hat{\omega}]^2(1-\cos\theta)\}x\end{aligned}$$

Given $\hat{\omega}$ and θ, what is $R \in \mathbb{SO}(3)$?

- First question: Is there a **unique** parametrization?
  - No:

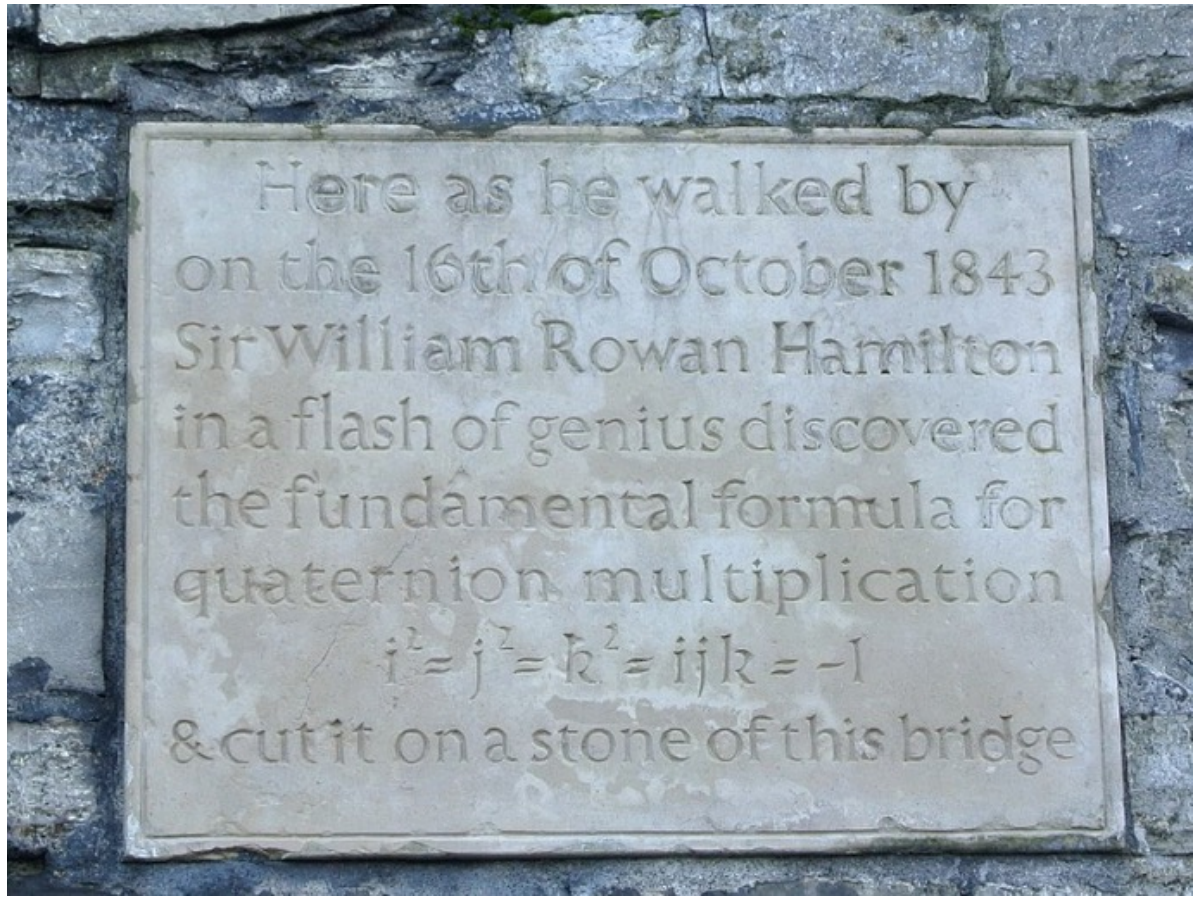
    1. $(\hat{\omega}, \theta)$ and $(-\hat{\omega}, -\theta)$ give the same rotation

    2. when $R = I$, $\theta = 0$ and $\hat{\omega}$ can be arbitrary

    3. $(\hat{\omega}, \pi)$ and $(-\hat{\omega}, \pi)$ give the same rotation $(\mathrm{tr}(R) = -1)$
- If we restrict $\theta \in (0, \pi)$, a unique parameterization exists:
  - $\theta = \arccos \frac{1}{2}[\mathrm{tr}(R) - 1], \quad [\hat{\omega}] = \frac{1}{2\sin\theta}(R - R^T)$

Quaternion

Note: In this section, $\vec{x} \in \mathbb{R}^3$ and $q \in \mathbb{R}^4$

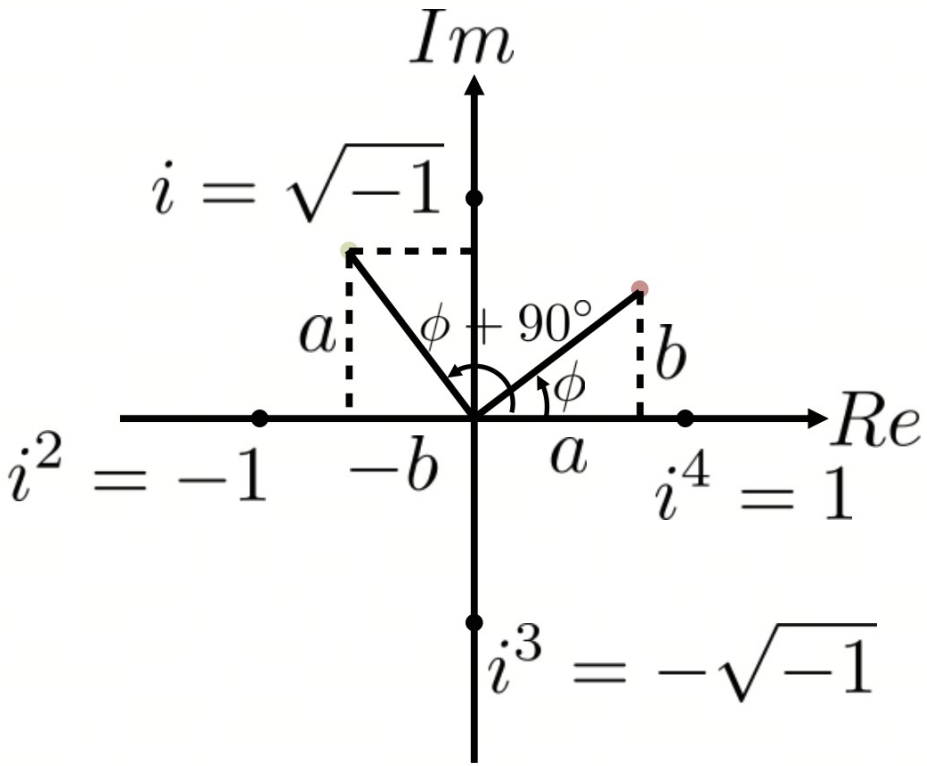


Quaternions were discovered by Sir William Rowan Hamilton in 1843

Quaternion is a "Number"

- A brief review of unit-magnitude complex numbers

$$a + ib \qquad\qquad a^2 + b^2 = 1$$



$$b = \sin\phi \qquad a = \cos\phi$$

$$\phi = \operatorname{atan2}\left(\frac{b}{a}\right)$$

Multiplying by i causes a 90° rotation in the plane.

$$(a + ib)(i) = -b + ia$$

Quaternion is a "Number"

- Recall the complex number $a + b\mathbf{i}$
- Quaternion is a more generalized complex number:

$$q = w + x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$

  - w is the real part and $\vec{v} = (x, y, z)$ is the imaginary part
  - Imaginary: $\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = \mathbf{ijk} = -1$
  - anti-commutative : $\mathbf{ij} = \mathbf{k} = -\mathbf{ji}, \mathbf{jk} = \mathbf{i} = -\mathbf{kj}, \mathbf{ki} = \mathbf{j} = -\mathbf{ik}$

Properties of General Quaternions

- Vector form: $q = (w, \vec{v})$
- Product:
  - For $q_1 = (w_1, \vec{v}_1)$ and $q_2 = (w_2, \vec{v}_2)$,
    $$q_1 q_2 = (w_1 w_2 - \vec{v}_1^T \vec{v}_2, w_1 \vec{v}_2 + w_2 \vec{v}_1 + \vec{v}_1 \times \vec{v}_2)$$
  - Not commutable (note that $\vec{v}_1 \times \vec{v}_2 \neq \vec{v}_2 \times \vec{v}_1$)
- Conjugate: $q^* = (w, -\vec{v})$
- Norm: $\| q \|^2 = w^2 + \vec{v}^T \vec{v} = q q^* = q^* q$
- Inverse: $q^{-1} := \frac{q^*}{\|q\|^2}$

Unit Quaternion as Rotation

- A **unit** quaternion $\|\boldsymbol{q}\| = 1$ can represent a rotation
  - Four numbers plus one constraint → 3 DoF
- Geometrically, the shell of a 4D sphere

Build Rotation Quaternion

- Exponential coordinate → Quaternion:

$$q = [\cos(\theta/2), \sin(\theta/2)\,\hat{\omega}]$$

- Quaternion is very close to angle-axis representation!
- Exponential coordinate ← Quaternion: $\theta = 2\arccos(w)$,

$$\hat{\omega} = \begin{cases} \frac{1}{\sin(\theta/2)}\vec{v} & \theta \neq 0 \\ 0 & \theta = 0 \end{cases}$$

Unit Quaternion as Rotation

- Rotate a vector $\vec{x}$ by a quaternion q:
  1. Augment $\vec{x}$ to $x = (0, \vec{x})$
  2. $x' = qxq^{-1}$
- Compose rotations by quaternion:
  - $(q_2(q_1 x q_1^*)q_2^*)$: first rotate by q_1 and then by q_2
  - Since $(q_2(q_1 x q_1^*)q_2^*) = (q_2 q_1)x(q_1^* q_2^*)$, composing rotations is as simple as multiplying quaternions!

Proof

Theorem 1 *For any unit quaternion*

$$q = q_0 + \boldsymbol{q} = \cos\frac{\theta}{2} + \boldsymbol{u}\sin\frac{\theta}{2}, \tag{4}$$

and for any vector $\boldsymbol{v} \in \mathbb{R}^3$ *the action of the operator*

$$L_q(\boldsymbol{v}) = q\boldsymbol{v}q^*$$

on $\boldsymbol{v}$ *is equivalent to a rotation of the vector through an angle* θ *about* $\boldsymbol{u}$ *as the axis of rotation.*

Proof Given a vector $\boldsymbol{v} \in \mathbb{R}^3$, we decompose it as $\boldsymbol{v} = \boldsymbol{a} + \boldsymbol{n}$, where $\boldsymbol{a}$ is the component along the vector $\boldsymbol{q}$ and $\boldsymbol{n}$ is the component normal to $\boldsymbol{q}$. Then we show that under the operator L_q, $\boldsymbol{a}$ is invariant, while $\boldsymbol{n}$ is rotated about $\boldsymbol{q}$ through an angle θ. Since the operator is linear, this shows that the image $q\boldsymbol{v}q^*$ is indeed interpreted as a rotation of $\boldsymbol{v}$ about $\boldsymbol{q}$ through an angle θ.

We know from an early reasoning that $\boldsymbol{a}$ is invariant under L_q. So let us focus on the effect of L_q on the orthogonal component $\boldsymbol{n}$. We have

$$\begin{aligned} L_q(\boldsymbol{n}) &= (q_0^2 - \|\boldsymbol{q}\|^2)\boldsymbol{n} + 2(\boldsymbol{q}\cdot\boldsymbol{n})\boldsymbol{q} + 2q_0(\boldsymbol{q}\times\boldsymbol{n}) \\ &= (q_0^2 - \|\boldsymbol{q}\|^2)\boldsymbol{n} + 2q_0(\boldsymbol{q}\times\boldsymbol{n}) \\ &= (q_0^2 - \|\boldsymbol{q}\|^2)\boldsymbol{n} + 2q_0\|\boldsymbol{q}\|(\boldsymbol{u}\times\boldsymbol{n}), \end{aligned}$$

where in the last step above we introduced $\boldsymbol{u} = \boldsymbol{q}/\|\boldsymbol{q}\|$. Denote $\boldsymbol{n}_\perp = \boldsymbol{u}\times\boldsymbol{n}$. So the last equation becomes

$$L_q(\boldsymbol{n}) = (q_0^2 - \|\boldsymbol{q}\|^2)\boldsymbol{n} + 2q_0\|\boldsymbol{q}\|\boldsymbol{n}_\perp. \tag{5}$$

Note that $\boldsymbol{n}_\perp$ and $\boldsymbol{n}$ have the same length:

$$\|\boldsymbol{n}_\perp\| = \|\boldsymbol{n}\times\boldsymbol{u}\| = \|\boldsymbol{n}\|\cdot\|\boldsymbol{u}\|\sin\frac{\pi}{2} = \|\boldsymbol{n}\|.$$

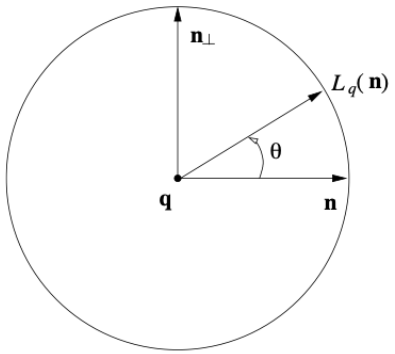
Proof credit: Stanford CS348A (link)

Proof (Cont'd)

Finally, we rewrite (5) into the form

$$\begin{aligned} L_q(\boldsymbol{n}) &= \left(\cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2}\right)\boldsymbol{n} + \left(2\cos\frac{\theta}{2}\sin\frac{\theta}{2}\right)\boldsymbol{n}_\perp \\ &= \cos\theta\boldsymbol{n} + \sin\theta\boldsymbol{n}_\perp. \end{aligned}$$

Namely, the resulting vector is a rotation of $\boldsymbol{n}$ through an angle θ in the plane defined by $\boldsymbol{n}$ and $\boldsymbol{n}_\perp$. See the figure below. This vector is clearly orthogonal to the rotation axis.



We substitute the unit quaternion form (4) into (3) to obtain the resulting vector from rotating a vector $\boldsymbol{v}$ about the axis $\boldsymbol{u}$ through θ:

$$\begin{aligned} L_p(\boldsymbol{v}) &= \left(\cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2}\right)\boldsymbol{v} + 2\left(\boldsymbol{u}\sin\frac{\theta}{2}\cdot\boldsymbol{v}\right)\boldsymbol{u}\sin\frac{\theta}{2} + 2\cos\frac{\theta}{2}\left(\boldsymbol{u}\sin\frac{\theta}{2}\times\boldsymbol{v}\right) \\ &= \cos\theta\cdot\boldsymbol{v} + (1-\cos\theta)(\boldsymbol{u}\cdot\boldsymbol{v})\boldsymbol{u} + \sin\theta\cdot(\boldsymbol{u}\times\boldsymbol{v}). \end{aligned} \tag{6}$$

$$\mathrm{Rot}(\mathbf{u},\theta)\mathbf{v} = \mathbf{v} + (1-\cos\theta)\mathbf{u}\times(\mathbf{u}\times\mathbf{v}) + (\sin\theta)\mathbf{u}\times\mathbf{v}$$

$$\because\ \mathbf{u}\times(\mathbf{u}\times\mathbf{v}) = (\mathbf{u}\cdot\mathbf{v})\mathbf{u} - (\mathbf{u}\cdot\mathbf{u})\mathbf{v} = (\mathbf{u}\cdot\mathbf{v})\mathbf{u} - \mathbf{v}$$

$$\therefore\ \mathrm{Rot}(\mathbf{u},\theta)\mathbf{v} = \mathbf{v}\cos\theta + (1-\cos\theta)(\mathbf{u}\cdot\mathbf{v})\mathbf{u} + (\sin\theta)\mathbf{u}\times\mathbf{v} = L_p(\mathbf{v})$$

Unit Quaternion as Rotation

- Rotate a vector $\vec{x}$ by a quaternion q:
  1. Augment $\vec{x}$ to $x = (0, \vec{x})$
  2. $x' = qxq^{-1}$
- Compose rotations by quaternion:
  - $(q_2(q_1 x q_1^*)q_2^*)$: first rotate by q_1 and then by q_2
  - Since $(q_2(q_1 x q_1^*)q_2^*) = (q_2 q_1)x(q_1^* q_2^*)$, composing rotations is as simple as multiplying quaternions!
- Each rotation corresponds to two quaternions ("double-covering")

Quaternion vs. Rotation Matrix

Storage:

- A quaternion: 4 floating-point values (x, y, z, w),
- A rotation matrix: 9 values (3×3 matrix).

Multiplication:

- Quaternion : 16 multiplications and 12 additions per operation
- Rotation matrix: 27 multiplications and 18 additions

Numerical stability:

- Rotation matrices can accumulate numerical errors over successive operations (e.g., due to floating-point precision limits), potentially violating orthogonality.
- Quaternions, when normalized, maintain unit magnitude.

More about Quaternion

- Quaternion is computationally cheap:
  - Internal representation of Physical Engine and Robot
- Pay attention to convention (w, x, y, z) or (x, y, z, w)
  - (w, x, y, z): SAPIEN, transforms3d, Eigen, blender, MuJoCo, V-Rep
  - (x, y, z, w): ROS, PhysX, PyBullet

```
template<typename _Scalar>
Eigen::Quaternion< _Scalar >::Quaternion ( Scalar w,
                                           Scalar x,
                                           Scalar y,
                                           Scalar z
                                         )        [inline]
```

Constructs and initializes the quaternion $ w+xi+yj+zk $ from its four coefficients *w*, *x*, *y* and *z*.

Warning:
Note the order of the arguments: the real *w* coefficient first, while internally the coefficients are stored in the following order: [x, y, z, w]

Definition at line **108** of file **Eigen2Support/Geometry/Quaternion.h**.

Conversion between Quaternion and Axis-Angle

- Axis-angle → Quaternion:

$$q = [\cos(\theta/2), \sin(\theta/2)\ \hat{\omega}]$$

- Quaternion is very close to angle-axis representation!
- Axis-angle ← Quaternion: $\theta = 2\arccos(w)$,

$$\hat{\omega} = \begin{cases} \dfrac{1}{\sin(\theta/2)}\vec{v} & \theta \neq 0 \\ 0 & \theta = 0 \end{cases}$$

Conversion between Quaternion and Rotation Matrix

- Rotation Matrix <- Quaternion
  - Since we have $x' = qxq^{-1}$, we can calculate the rotation matrix as:
  - $R(q) = \begin{bmatrix} 1-2y^2-2z^2 & 2xy-2zw & 2xz+2yw \\ 2xy+2zw & 1-2x^2-2z^2 & 2yz-2xw \\ 2xz-2yw & 2yz+2xw & 1-2x^2-2y^2 \end{bmatrix}$

- Quaternion <- Rotation Matrix
  - As $tr(R) = 3 - 4(x^2+y^2+z^2) = 4w^2 - 1$ ($tr(R)$ is the trace of R)
  - We can calculate the quaternion as $w = \frac{\sqrt{tr(R)+1}}{2}, x = \frac{R_{32}-R_{23}}{4w}, y = \frac{R_{13}-R_{31}}{4w}, z = \frac{R_{21}-R_{12}}{4w}$

Distance between Quaternions

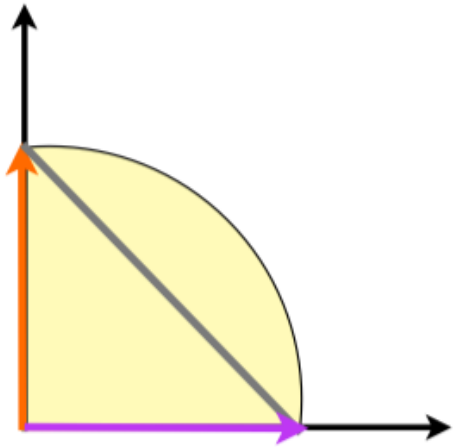
- The distance between quaternions (q_1, q_2)?
  - The distance in $\mathbb{S}^3$ or the angle between (q_1, q_2): $< p, q >= \arccos(p \cdot q)$
  - The distance between the corresponding rotations
    - $dist(p, q) = 2\arccos(|p \cdot q|)$ (prove by yourself)

$$= 2min\{< p, q >, < p, -q >\}$$

- The distance between rotations (R_1, R_2) is linear to the on-sphere distance between $q(R_1)$ and $q(R_2)$.

Interpolating between quaternions

- **Why not linear interpolation?**
- Need to be normalized
- Does not have a constant rate of rotation



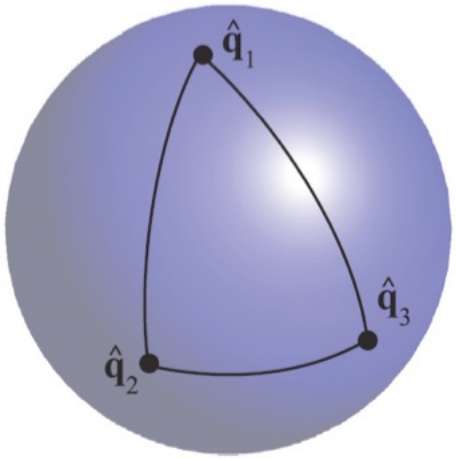
$$\frac{(1-\alpha)x + \alpha y}{||(1-\alpha)x + \alpha y||}$$

Slides credit: Cornell CS4620 Fall 2015

Spherical Linear Interpolation

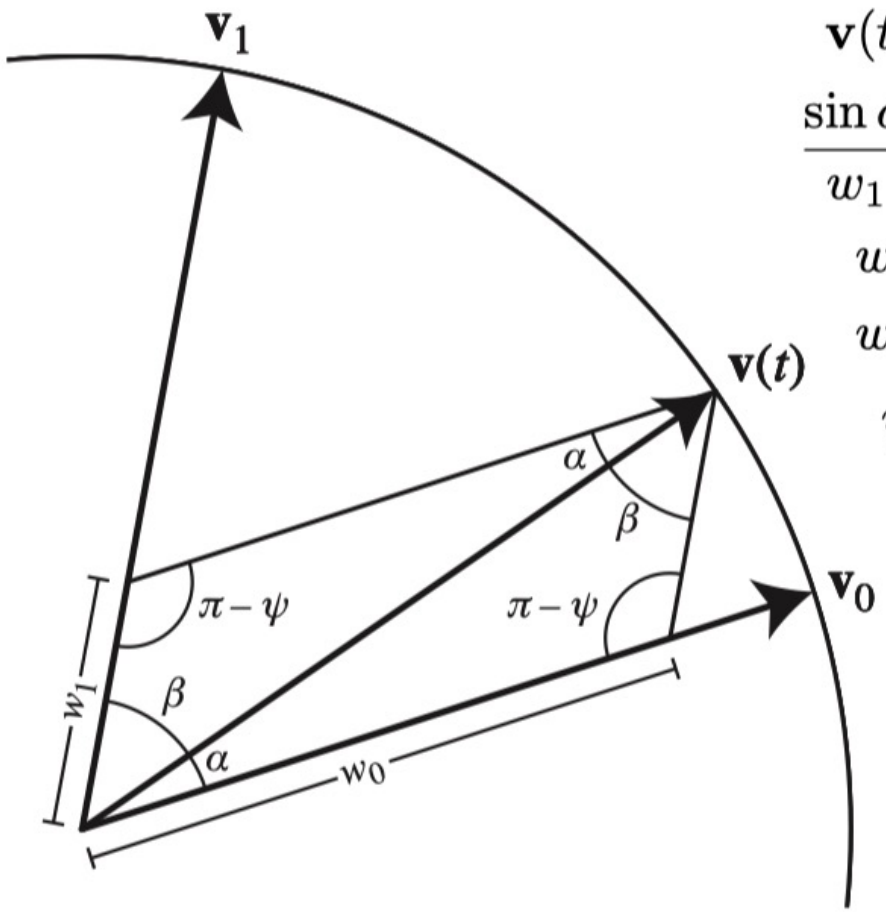
Intuitive interpolation between different orientations

- Nicely represented through quaternions
- **Useful for animation and motion planning**
- Given two quaternions, interpolate between them
- Shortest path between two points on sphere
  - Geodesic, on Great Circle



Slides credit: Cornell CS4620 Fall 2015

Spherical linear interpolation ("slerp")



$$\alpha + \beta = \psi$$

$$\mathbf{v}(t) = w_0 \mathbf{v}_0 + w_1 \mathbf{v}_1$$

$$\frac{\sin\alpha}{w_1} = \frac{\sin\beta}{w_0} = \frac{\sin(\pi - \psi)}{1} = \sin\psi$$

$$w_0 = \sin\beta / \sin\psi$$

$$w_1 = \sin\alpha / \sin\psi$$

$$\psi = \cos^{-1}(\mathbf{v}_0 \cdot \mathbf{v}_1)$$

Slides credit: Cornell CS4620 Fall 2015

Quaternion Interpolation

- Spherical linear interpolation (SLERP) naturally works in any dimension
- Traverses a great arc on the sphere of unit quaternions
  - Uniform angular rotation velocity about a fixed axis

$$\psi = \cos^{-1}(q_0 \cdot q_1)$$

$$q(t) = \frac{q_0 \sin(1-t)\psi + q_1 \sin t\psi}{\sin \psi}$$

Slides credit: Cornell CS4620 Fall 2015

Practical issues

- When the angle between q_1 and q_2 gets close to zero, use small angle approximation
  - degenerate to linear interpolation
- When angle close to 180, there is no shortest geodesic, but can pick one
- q is same rotation as $-q$
  - if the angle between q_1 and $q_2 < \pi/2$, slerp between them;
  - else, slerp between q_1 and q_2.

Slides credit: Cornell CS4620 Fall 2015

Uniform Sampling in SO(3)

- To uniformly sample a rotation from $\mathbb{SO}(3)$ is the same to uniformly sample a unit quaternion from $\mathbb{S}(3)$
  - Why? Because the distance between two rotations is linear to the on-sphere distance between the two corresponding quaternions.

- How to uniformly sample $\mathbb{S}(3)$**?**
  - The simplest way is to random sample a variable from $\mathcal{N}(0, I_{4\times4})$ and normalize it to unit length.
    - Why? Because standard normal distribution is isotropic.
  - Then convert this quaternion to a rotation if you want.

Also refer to https://niepp.github.io/2021/12/09/uniform-sampling-on-sphere.html for analytical solution

Resources

- In Python, you could use the transforms3d library
- For differentiable transformations, you can play with is “Kornia’’, but use with cautious to its numerical properties
- “ceres” is a C++ library that is quite useful

Why Quaternions?

- **Very useful and popular in practice**
  - Fast, few operations, not redundant (4D compact parameterization, and efficient to compute)
  - Numerically stable for incremental changes
  - Composes rotations nicely

- **Convert to matrices at the end**

Summary of Rotation Representations

	Inverse?	Composing?	Any local movement in SO(3) can be achieved by local movement in the domain?
Rotation Matrix	✓	✓	N/A
Euler Angle	Complicated	Complicated	No
Angle-axis	✓	Complicated	?
Quaternion	✓	✓	✓

? means no singularity with single exceptions

Summary of Rotation Representations

- **It is quite often that, we use**
  - rotation matrices to define concepts
  - Euler angles to visualize rotations
  - angle-axis representation to visualize rotations and calculate derivatives
  - quaternion to write fast codes

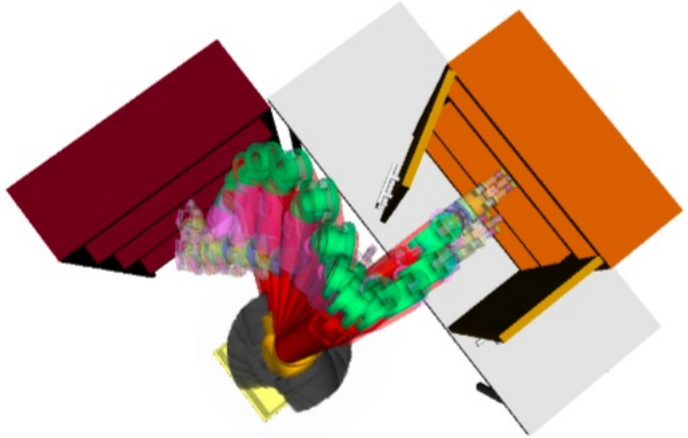
Refer to CVPR'19 paper, On the Continuity of Rotation Representations, and a recent TPAMI paper from my lab, Towards Robust Probabilistic Modeling on SO(3) via Rotation Laplace Distribution, to know more about rotation representations for learning purposes.

Motion Planning

Configuration Space

- Configuration space ($\mathcal{C}$-space) is a subset of $\mathbb{R}^n$ containing all possible states of the system(state space in RL).
- $\mathcal{C}_{free} \subseteq \mathcal{C}$ contains all valid states.
- $\mathcal{C}_{obs} \subseteq \mathcal{C}$ represents obstacles.
- Examples:
  - All valid poses of a robot.
  - All valid joint values of a robot.
  - ...

Examples



- *Ratliff N, Zucker M, Bagnell J A, et al. CHOMP: Gradient optimization techniques for efficient motion planning, ICRA 2009*
- *Schulman, John, et al. Finding Locally Optimal, Collision-Free Trajectories with Sequential Convex Optimization, RSS 2013*

Motion Planning

- Problem:
  - Given a configuration space $\mathcal{C}_{free}$
  - Given start state q_{start} and goal state q_{goal} in $\mathcal{C}_{free}$
  - Calculate a sequence of actions that leads from start to goal
- Challenge:
  - Need to avoid obstacles
  - Long planning horizon
  - High-dimensional planning space

Motion Planning

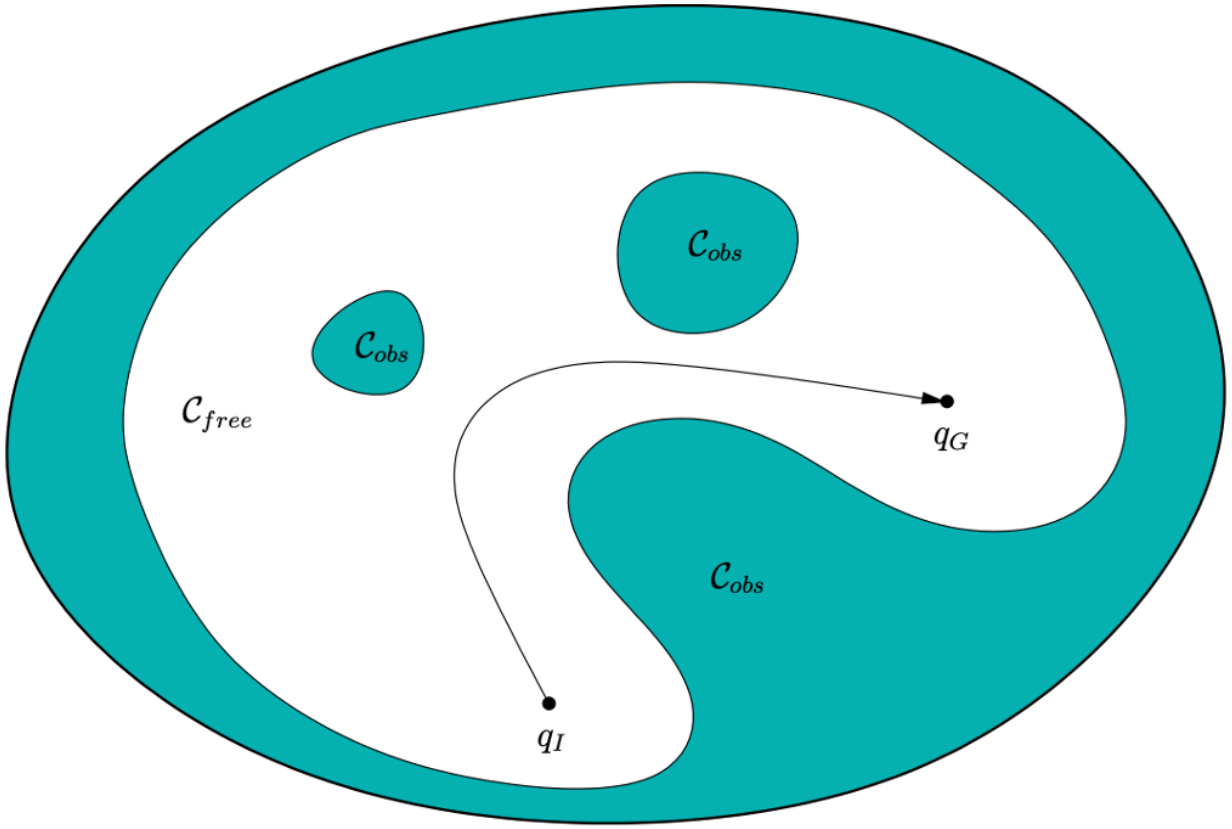
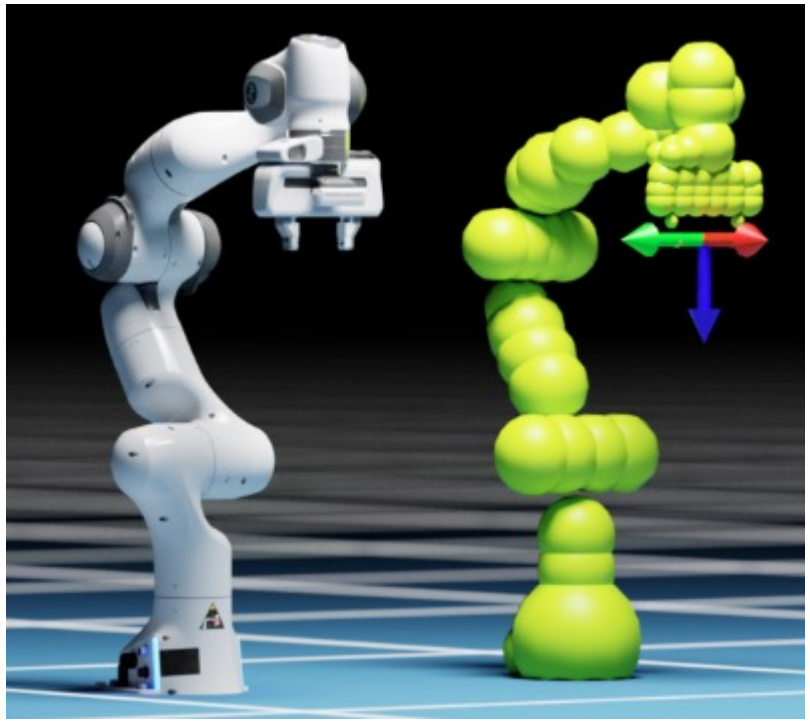


Figure 4.11: The basic motion planning problem is conceptually very simple using C-space ideas. The task is to find a path from q_I to q_G in $\mathcal{C}_{free}$. The entire blob represents $\mathcal{C} = \mathcal{C}_{free} \cup \mathcal{C}_{obs}$.

Collision modeling of robot

- Collision mesh of each link is defined in URDF
- Use forward-kinematics to get robot collision mesh from q
- Run collision check to see if q is in C_{free}



Robot representation in cuRobo

Next week: Lecture 4,
Robotics III